

Explanation of Code and DataSet and Inferences from the Output

DataSet Explanation

1. Random

- A categorical variable indicating the experimental grouping or randomization block in which the plants were grown.
 - Used to control for variability not related to the treatments — helps ensure fair comparison.
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2. Class

- Represents the category or treatment group of the plants, such as different cultivars, genotypes, or experimental treatments.
 - Distinguishes the different plant groups being compared.
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3. ACHP — Average Chlorophyll Content per Plant (inferred)

- Likely measures the average chlorophyll level, indicating the photosynthetic pigment concentration.
 - Higher values typically indicate healthier, more photosynthetically active plants.
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4. PHR — Plant Height Rate or Plant Height Relative

- Measures growth rate or relative height of the plant, possibly in centimeters.
 - Reflects vertical growth vigor, an important agronomic trait.
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5. AWWGV — Average Weekly Water Gain Volume (inferred)

- Amount of water gained or absorbed by the plant weekly, perhaps measured in ml or liters.
 - Indicates water uptake efficiency.
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6. ALAP — Average Leaf Area per Plant

- Total leaf area of the plant, typically measured in cm^2 or m^2 .
 - Larger leaf area generally means more surface for photosynthesis.
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7. ANPL — Average Number of Leaves per Plant

- Total leaf count per plant.
 - More leaves can mean higher photosynthetic capacity but might also indicate resource allocation trade-offs.
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8. ARD — Average Root Depth

- Measurement of root depth, probably in cm.
 - Deeper roots can access more water and nutrients.
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9. ADWR — Average Daily Water Requirement (inferred)

- Amount of water the plant needs daily, in ml or liters.
 - Useful for irrigation scheduling.
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10. PDMVG — Plant Dry Matter Vegetative Growth (inferred)

- Dry biomass accumulation during the vegetative growth phase.
 - Reflects plant growth efficiency and resource allocation.
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11. ARL — Average Root Length

- Measurement of root length in cm.
 - Important for nutrient absorption.
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12. AWWR — Average Weekly Water Requirement

- Water needed per week, useful for planning irrigation.
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13. ADWV — Average Daily Water Volume

- Volume of water provided or consumed daily.
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14. PDMRG — Plant Dry Matter Reproductive Growth (inferred)

- Dry biomass accumulated during reproductive phase (flowers, fruits).
 - Indicator of yield potential.
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Notes on Inference:

- The exact definitions should ideally come from your project documentation or dataset metadata.
- Units may vary (cm, ml, g, cm², etc.).
- Some acronyms may be specific to your institution or dataset.

Explanation of the Code

1. Importing Necessary Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

- numpy as np: A library for numerical operations, arrays, and mathematical functions.
 - pandas as pd: Library for handling tabular data in DataFrames, enabling easy data manipulation.
 - matplotlib.pyplot as plt: A plotting library for creating static, animated, and interactive visualizations like bar charts.
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2. Reading the Dataset

```
data = '/Users/dhruvkarande/Documents/Data Analytics Project /Green House Plant Growth/
Greenhouse Plant Growth Metrics.csv'
df = pd.read_csv(data)
```

- data: String storing the absolute path to the CSV file.
 - pd.read_csv(data): Reads CSV file from the path and returns a pandas DataFrame, stored in df.
 - df now holds all the tabular data from your CSV.
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3. Previewing the Dataset

```
print("First 5 rows:")
print(df.head())
```

- df.tail(): Returns last 5 rows, useful to confirm data completeness at the bottom.
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4. Inspecting Dataset Structure

```
print(df.columns)
```

- Prints a list of all column names in the dataset.

```
print(df["Random"].unique())
print(df["Class"].unique())
```

- df["Random"].unique() prints all unique values in the "Random" column.
 - Similarly for "Class".
 - Helps understand how many different groups/categories exist.
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5. Cleaning the Data

```
df = df.dropna()
```

- Removes any rows containing missing/null values (NaNs).

- Ensures subsequent analysis uses only complete data.

```
net_null = df.notnull().sum()
print(net_null)
```

- `df.notnull()` creates a boolean DataFrame showing True for non-null entries.
 - `.sum()` adds up True values per column — effectively counts non-missing values.
 - Prints number of valid entries per column post-cleaning.
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6. Convert Categorical Variables to Numeric

```
df['Random'] = df['Random'].astype('category').cat.codes
df['Class'] = df['Class'].astype('category').cat.codes
```

- Converts the "Random" and "Class" columns to categorical dtype.
 - `.cat.codes` assigns integer codes starting from 0 for each category.
 - Essential for numerical operations (correlation, plotting).
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7. Define Columns for Correlation

```
set = ['Random', 'ACHP', 'PHR', 'AWWGV', 'ALAP', 'ANPL', 'ARD', 'ADWR', 'PDMVG', 'ARL',
       'AWWR', 'ADWV', 'PDMRG', 'Class']
```

- Defines a list of column names to calculate correlation between.
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8. Calculate Correlation Matrix

```
corr_matrix = df[set].corr()
```

- Calculates pairwise Pearson correlation coefficients for columns in set.
- `.corr()` computes how strongly columns are linearly related.

```
print('Correlation Matrix')
print(corr_matrix.to_string())
```

- Prints the correlation matrix in a readable string format.
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9. Plotting Mean Values for Each Class-Random Group

a. Import plotting and math libraries (again, redundant but safe):

```
import matplotlib.pyplot as plt
from math import ceil
```

- `matplotlib.pyplot` for plotting.
 - `ceil` for ceiling function (not used).
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b. Calculate unique groups count:

```
num_classes = df['Class'].nunique()
num_randoms = df['Random'].nunique()
```

- Counts unique Class values — rows of subplot grid.
 - Counts unique Random values — columns of subplot grid.
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c. Create subplot grid:

```
fig, axes = plt.subplots(num_classes, num_randoms, figsize=(num_randoms * 6, num_classes * 4))
fig.suptitle("Mean Value Bar Charts for Each Class-Random Combination", fontsize=16)
```

- plt.subplots creates a grid of plots with dimensions [num_classes x num_randoms].
 - figsize scales figure size based on grid.
 - fig.suptitle adds a main title over all subplots.
-

d. Loop through all class-random pairs:

```
for class_val in range(num_classes):
    for random_val in range(num_randoms):
```

- Nested loops over class and random indices.

```
    filtered_df = df[(df['Class'] == class_val) & (df['Random'] == random_val)]
    if filtered_df.empty:
        continue
```

- Filters rows where Class == class_val and Random == random_val.
 - Skips iteration if no data.
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e. Calculate mean of features for filtered data:

```
mean_vals = filtered_df.drop(columns=['Class', 'Random']).mean()
```

- Drops grouping columns from mean calculation.
 - Computes mean for all other columns.
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f. Plot bar chart in the correct subplot:

```
row = class_val
mean_vals.plot(kind='bar', ax=axes[row][random_val], color='skyblue', edgecolor='black')
```

- axes[row][random_val] targets subplot position.
- Plots a bar chart with skyblue fill and black edges.

```
axes[row][random_val].set_title(f'Bar: Class={class_val}, Random={random_val}', fontsize=10)
```

- Adds a title specifying class and random group.

```
axes[row][random_val].tick_params(axis='x', labelrotation=45)
```

- Rotates x-axis labels for readability.

g. Adjust layout and display all plots:

```
plt.tight_layout(rect=[0, 0.03, 1, 0.97])  
plt.show()
```

- Adjusts subplot spacing to prevent overlap.
- Displays the entire figure with all subplots.

Inference From the Output

1. Dominance of ALAP:

- The feature ALAP (Average Leaf Area per Plant) consistently shows much higher mean values than other parameters across all classes and random groups, indicating it is on a larger scale or more variable.

2. Relative Scale Differences:

- Other parameters (ACHP, PHR, ANPL, etc.) have much smaller mean values compared to ALAP, which visually compresses their differences in the bar charts.

3. Consistency Across Classes:

- The mean value profiles for each feature are generally consistent across different Class groups, suggesting similar growth patterns or treatment effects across classes.

4. Minimal Variation Across Random Groups:

- Within each class, the differences in mean values across Random groups (0, 1, 2) are subtle, implying the randomization factor has limited impact on the average feature values.

5. Potential Need for Normalization:

- Because of the large scale difference between ALAP and other features, applying feature scaling (e.g., normalization or standardization) would allow clearer visualization of relative changes in smaller-scale features.

6. Insight into Feature Importance:

- Features with consistently higher mean values might have greater influence on plant growth characteristics and warrant further detailed analysis.

7. Homogeneity in Growth Metrics:

- The relative uniformity across groups hints that treatments (represented by classes) may not drastically alter the measured plant growth metrics, or the dataset could require deeper segmentation for more nuanced insights.